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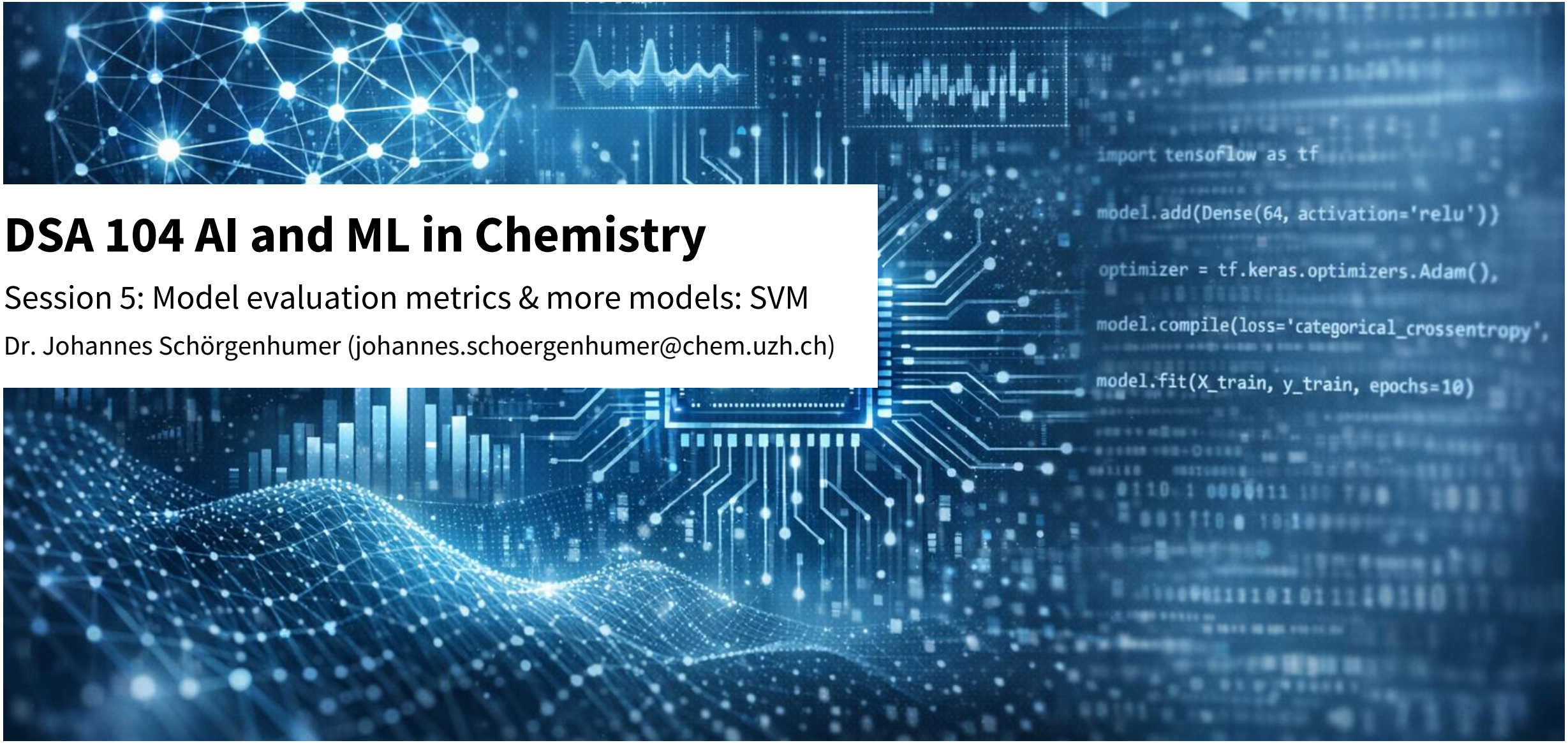


DSA 104 AI and ML in Chemistry

Session 5: Model evaluation metrics & more models: SVM

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

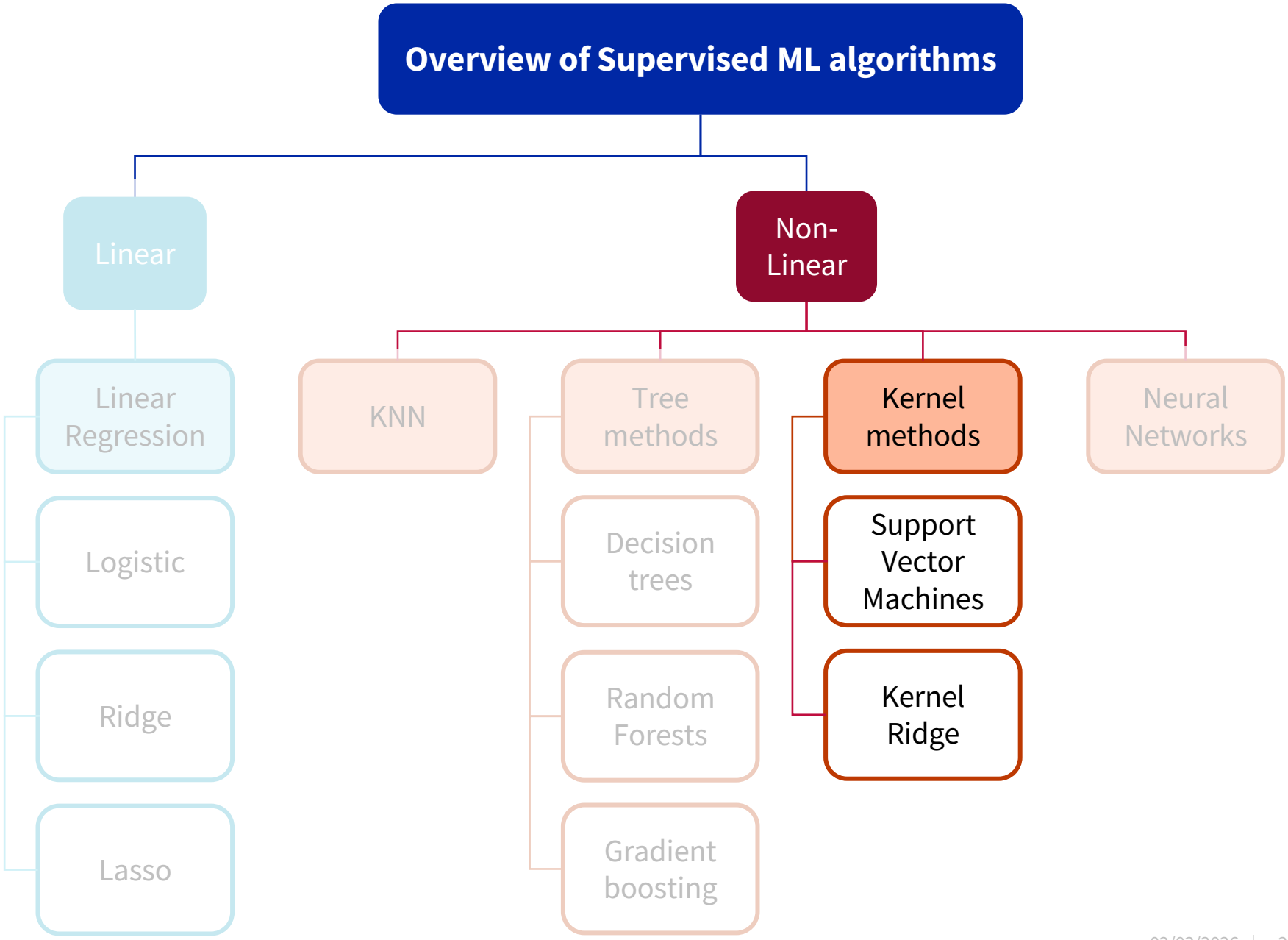
Lesson outline

Evaluation Recap

Data leakage

Metrics

Support Vector Machines



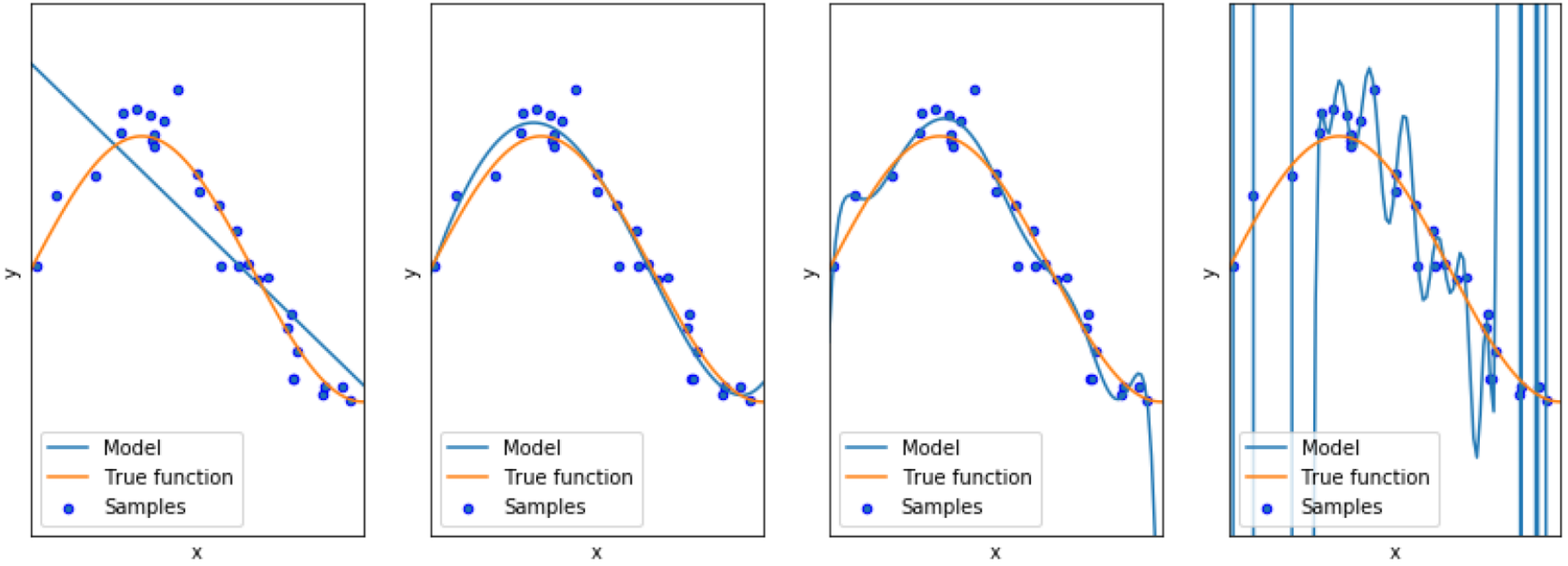
Evaluation: Recap

— Training error

— Generalisation error

— ERM

— Bias-variance tradeoff



Picture from A. Schörgenhumer, *Hands-on AI I*, Lecture materials, **2023**, JKU Linz.

Some remedies: Dataset Splitting Strategies

— Train/Test/Validation

— Cross validation: e.g. k-fold CV

Train	Train	Train	Train	Test
Train	Train	Train	Test	Train
Train	Train	Test	Train	Train
Train	Test	Train	Train	Train
Test	Train	Train	Train	Train

Fold = random split of shuffled data!

9	10	2	3	6	14	1	5	8	11	15	12	7	4	13

Other important CV methods:

— **Stratified k-fold**: response values close to equal in folds

— **Repeated k-fold**: repeat random splitting into k folds to average performance

Exhaustive CV methods: all possible splits explored

— **Leave one out (LOOCV)**: C_1^n (binomial coefficient)

— **Leave p out (LpOCV)**, e.g. p = 2 as almost unbiased method (but e.g. $C_2^{15} = \binom{15}{2} = 105$ possibilities…)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

⋮

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Computational Statistics & Data Analysis, **2011**, *55*, 1828-1844, https://doi.org/10.1016/j.csda.2010.11.018.

Risk: Data leakage

Data outside of the training data influences the model training!

Result: **optimistic performance** in validation **but poor generalisation**!

— Train-test contaminations, e.g. scaling on whole dataset, (partial) duplicated data in train and test splits

— Target leakage, e.g. model has access to features that encode the target

— Temporal leakage, e.g. random splits of time-series data

— CV leakage, e.g. reusing fitted transformers accross folds

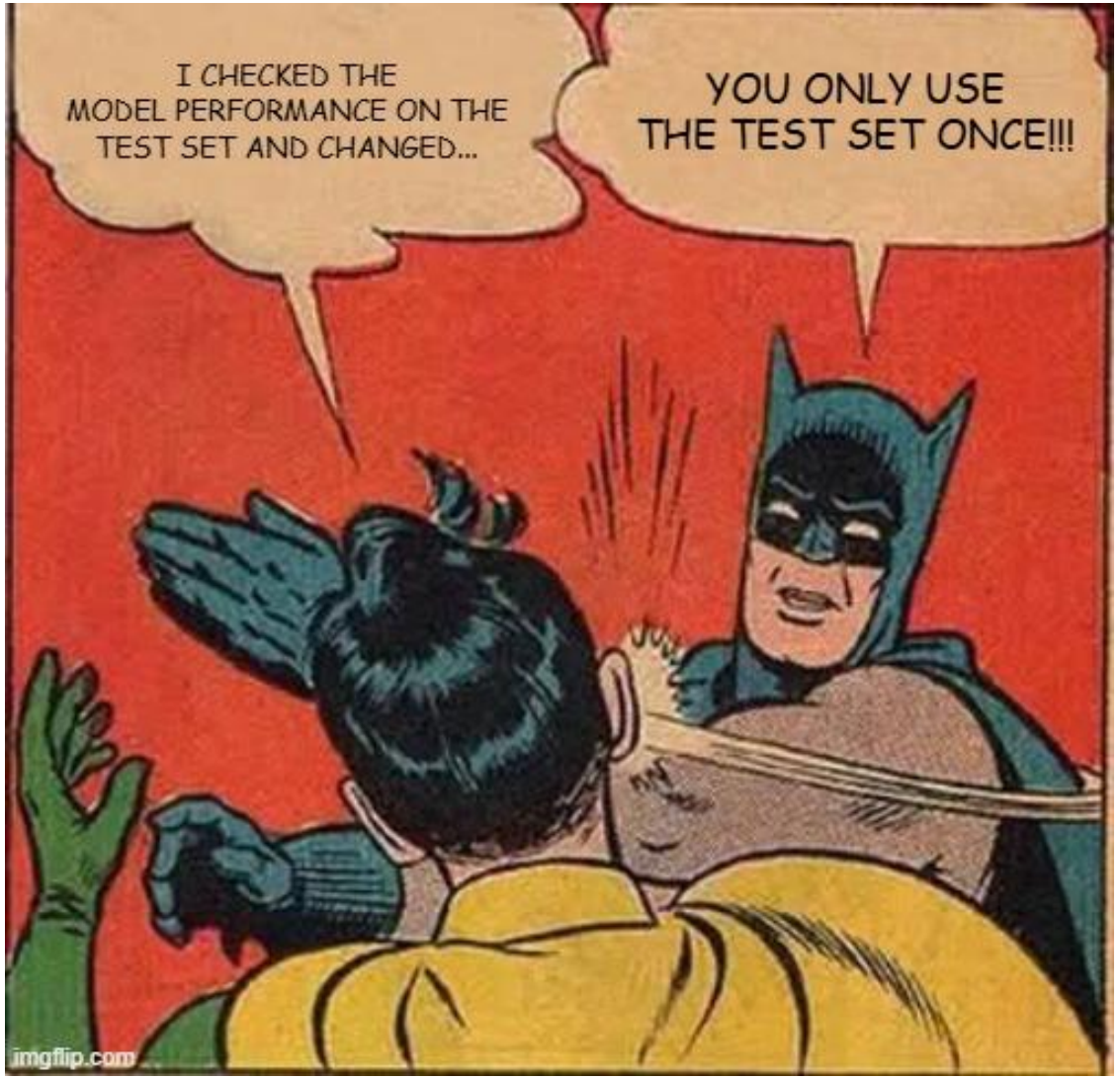
— Preprocessing leakage, e.g. transformation (PCA, normalisation, …) done before splitting

Good practice to avoid insufficient isolation:

— Keep strict separation: training, validation, testing!

— Perform transformations post-splitting

— Use pipelines to avoid “manual” mistakes

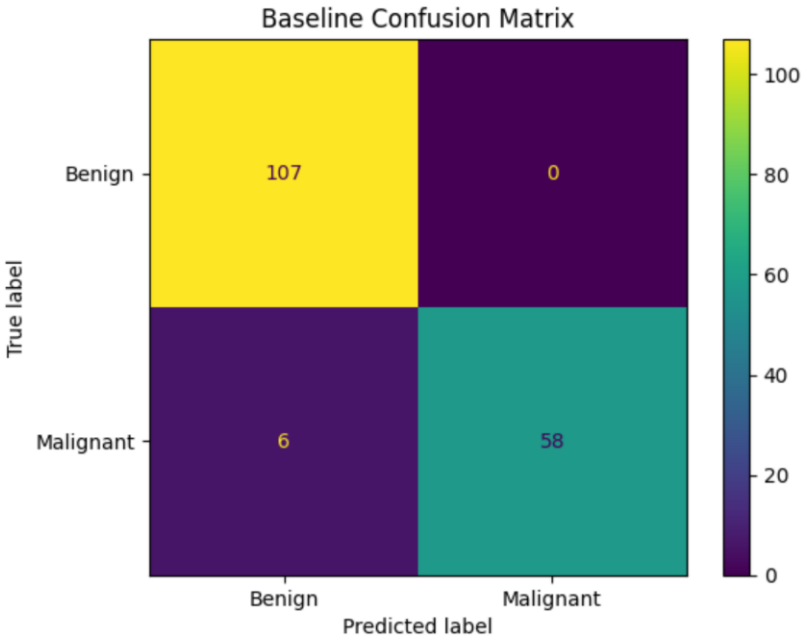


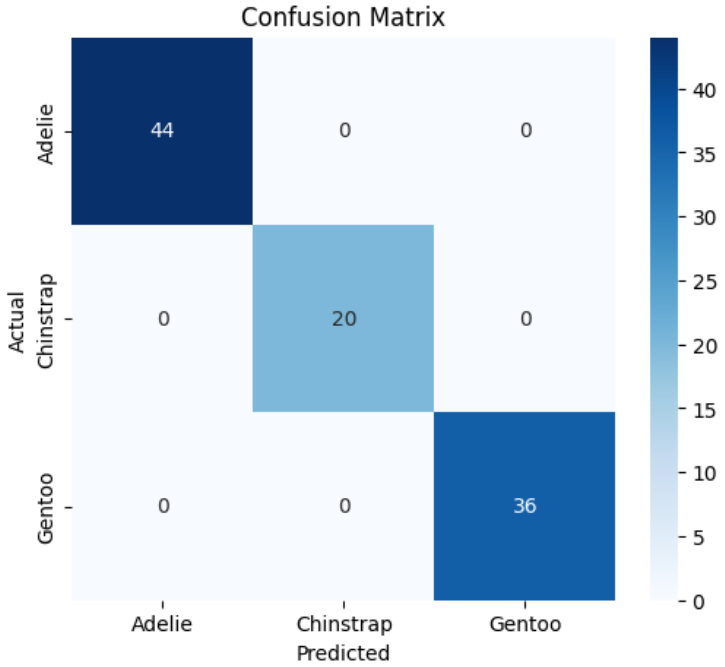
Classification Metrics

Confusion matrix:

— **Diagonal**: Correct predictions, e.g. true positives (**TP**) and true negatives (**TN**)

— **Off-diagonal**: Misclassifications, e.g. False positives (**FP**) and false negatives (**FN**)





Basis for further metrics, e.g. accuracy:

— Binary classification:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

— Generally (N x N):

$$Accuracy = \frac{\sum_i^N M_{ii}}{\sum_i^N \sum_j^N M_{ij}}$$

```
ConfusionMatrixDisplay.from_estimator(rf_default, X_test, y_test)
plt.title("Baseline Confusion Matrix")
plt.show()
```

```
cm = confusion_matrix(y_test, y_test_pred)

plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt="d",
            xticklabels=knn.classes_,
            yticklabels=knn.classes_,
            cmap="Blues")

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

Classification Metrics

Scores based on confusion matrix:

— **Accuracy:** How often the model is correct

$$\frac{TP + TN}{TP + TN + FP + FN}$$

— **Precision:** How many predicted positives were actually positive

$$\frac{TP}{TP + FP}$$

— **Recall (Sensitivity, TP rate (TPR):** How many actual positives did the model catch?

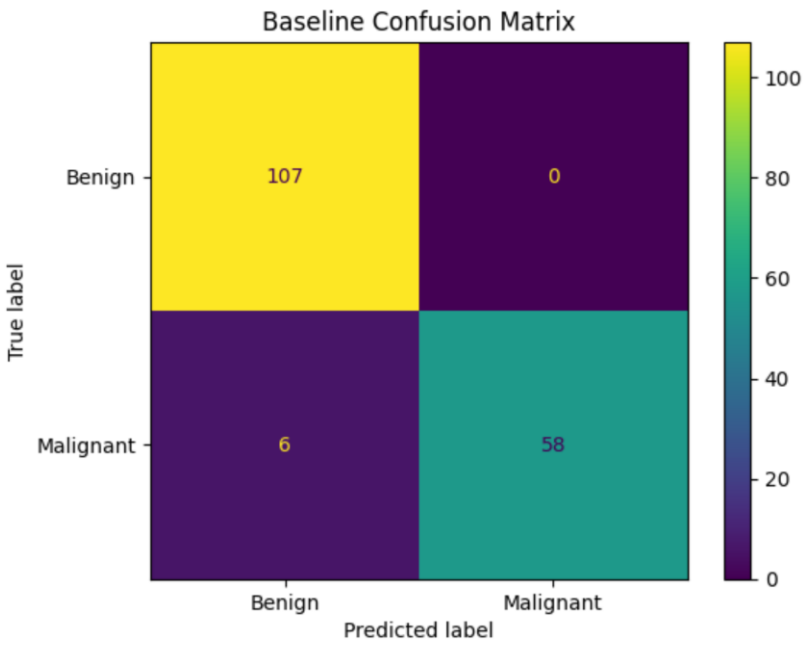
$$TPR = \frac{TP}{TP + FN}$$

— **Specificity (TNR):** How many actual negatives did the model catch?

$$TNR = \frac{TN}{TN + FP}$$

— **F1-Score:** Harmonic mean of precision & recall

$$2 \cdot \frac{precision \cdot recall}{precision + recall}$$



Alexander Elvers – own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=84255403

Classification Metrics: ROC & AUC

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN} = 1 - TNR$$

ROC (Receiver Operating Characteristics)

— ROC curve = plot of Recall (TPR) vs. FP rate (FPR = 1-Specificity (TNR))

— Essentially plots the trade-off **Sensitivity vs. Specificity** of a **Binary Classifier**

— **The higher the curve the better.**

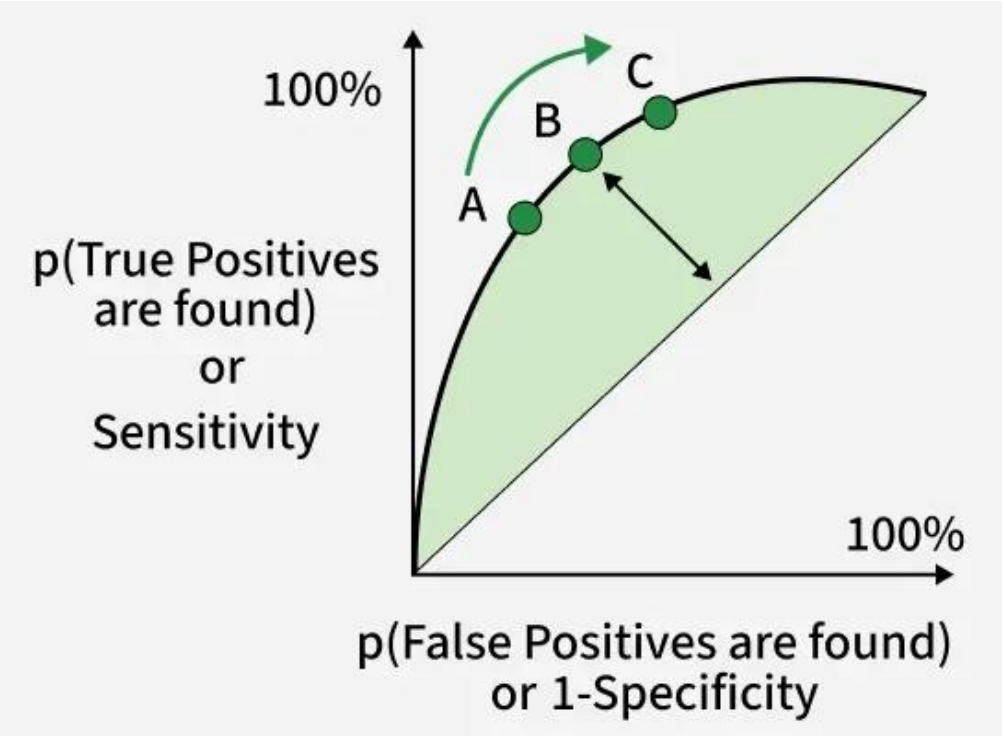
— **Each point represent threshold** (primary output of model = score, not class!)

ROC-AUC: area under the ROC curve

— 1 = entire graph, indicates perfect prediction

— 0.5 = diagonal = random guessing (50/50)

— Measures ranking **quality** (**independent of threshold**), not classification accuracy!



https://www.geeksforgeeks.org/machine-learning/auc-roc-curve/

Regression Metrics

Distinction according to how errors are penalised, interpretable they are, sensitive they are to outliers.

- **MAE (mean absolute error, L1)**
  - High interpretability (same unit)
  - Linear penalty for errors (More robust to outliers)

$$MAE = \frac{1}{n} \cdot \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

- **MSE (mean squared error, L2)**
  - Low interpretability (squared unit)
  - high penalty for outliers (less robust)

$$MSE = \frac{1}{n} \cdot \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- **RMSE (root MSE)**
  - Still high penalty for outliers (not more robust)
  - But more interpretable (same unit)

$$RMSE = \sqrt{MSE}$$

- **R^2**
  - Scale-free (no unit)
  - Error penalty indirectly high (via MSE)
  - Not an error magnitude!
  - Can be misleading!

$$R^2 = 1 - \frac{MSE}{Variance(y)}$$

ChatGPT's selection recommendation:

Use MAE when:
- You want robustness
- Errors should scale linearly
- Heavy-tailed noise expected

Use MSE/RMSE when:
- Large errors are unacceptable
- Gaussian noise assumption reasonable
- You want smooth optimisation

Use R^2 when:
- Comparing models on same dataset
- Communicating variance explained

Regression metrics

Consider the following minimal example:

[10, 10, 10, 10, 0] vs. predicted: [10, 10, 10, 10, 100]

— Calculate MAE, MSE, RMSE

— Discuss how these would affect training the model, i.e. what would the model try to do?

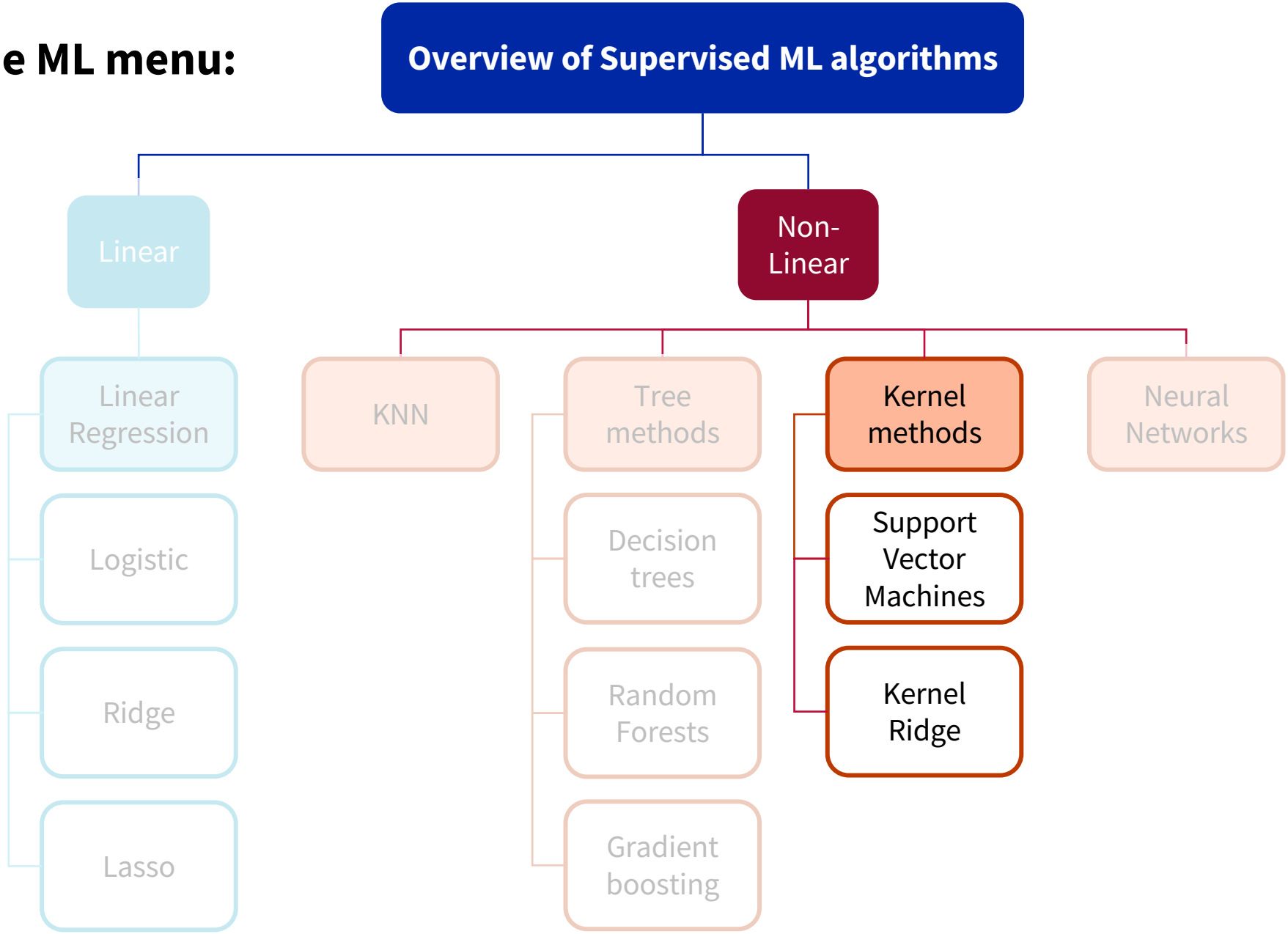
— Would you rather have one larger outlier and 4 perfect values, or 5 erroneous values?

— How is the noise distributed? Is MSE a valid metric in this case?

— Imagine a context. Is a 100-unit mistake catastrophic?

— When can you “ignore” a large error?

Next on the ML menu:



Support Vector Machines (SVM)

Supervised ML algorithm for classification (**SVC**) and regression (**SVR**)

SVC - main idea:

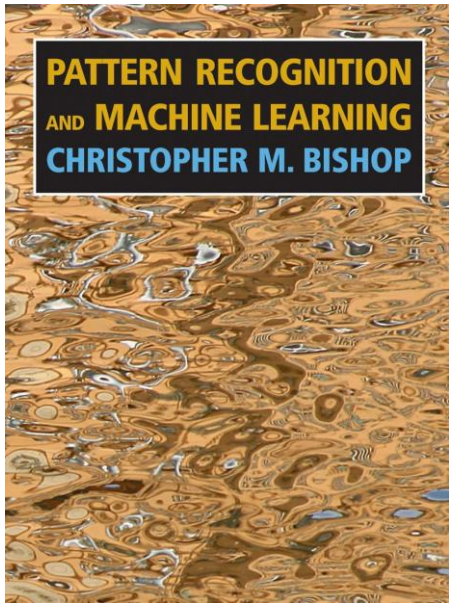
— Find the best possible boundary (***hyperplane***) that separates data points of different classes

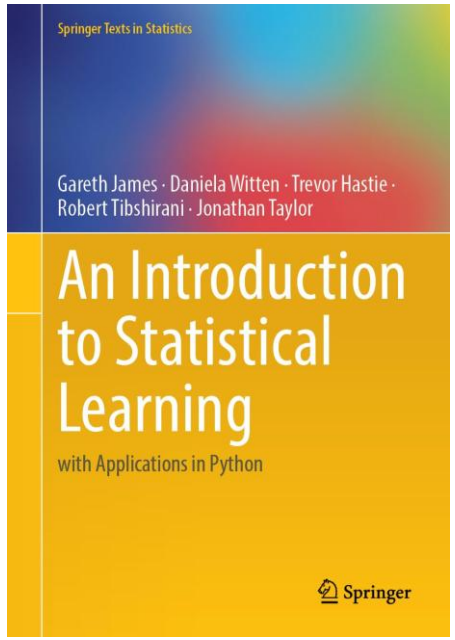
— Best boundary is the one that leaves the **largest margin** (i.e. distance) between itself and the closest data points.

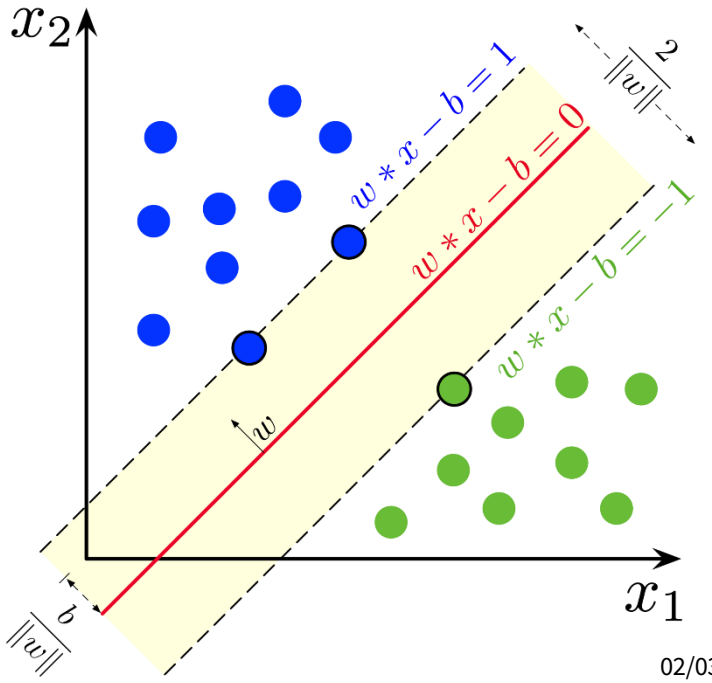
— Those closest points = ***support vectors***, “supporting” the position of the boundary

— SVMs work well with small–medium datasets, handle high dimensionality very well

— SVMs were dominant before deep learning, e.g. for image classification, semantic parsing (text), …







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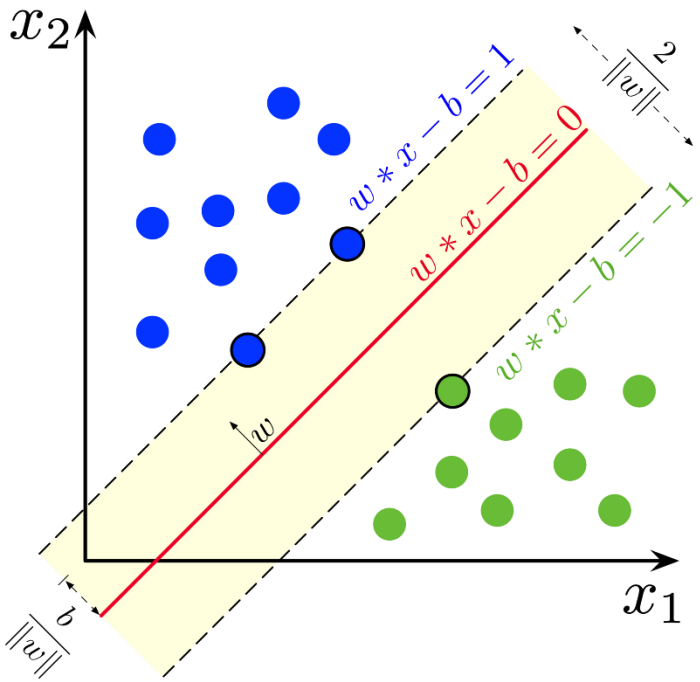
Support Vector Classifier (SVC): Linear kernel



Maximum margin classifier

Hyperplane described as $\left(w^T x_i - b\right) = 0$

…where w is a column vector transposed to deliver a scalar as dot product with x_i

…and the **bias** b describes the offset of the hyperplane with regards to the origin ($\frac{b}{\|w\|}$)

Note: bias is often defined with the opposite sign so that $\left(w^T x_i + b\right) = 0$

The SVM function $\mathrm{f}(x_i) = \mathrm{sgn}\left(w^T x_i - b\right)$ delivers 1 and -1 (**sign function**)

… thus the margin is: $\frac{2}{\|w\|}$

Perfectly separable = Hard margin SVM

To maximise the margin means to minimise $\|w\|$, or rather $\frac{1}{2}\|w\|^2$, subject to $y_i\left(w^T x_i - b\right) \geq 1$

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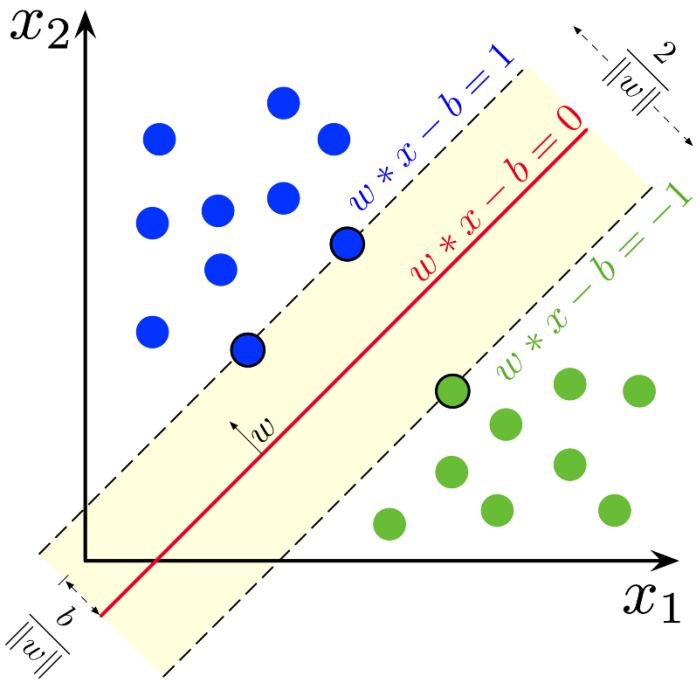
Support Vector Classifier (SVC): Linear kernel



Maximum margin classifier

More relevant: Soft-margin SVM
Hinge loss function:

$$\max(0, 1 - y_i(w^T x_i - b))$$

Hence minimise:

$$\|w\|^2 + C\frac{1}{n}\sum_{i=1}^{n}\max(0, 1 - y_i(w^T x_i - b)) = \|w\|^2 + C\sum_{i=1}^{n}\zeta_i$$

…with $\boldsymbol{\zeta_i}$ accounting for the hinge loss (**slack variables**)
…and **regularisation parameter *C*** as trade-off between margin size and penalty for points within

But: what if the data is not linearly separable?

Picture by Larhmam - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=73710028

Support Vector Classifier (SVC): The kernel trick

Instead of non-linear separation: transform slinear SVM into non-linear model!

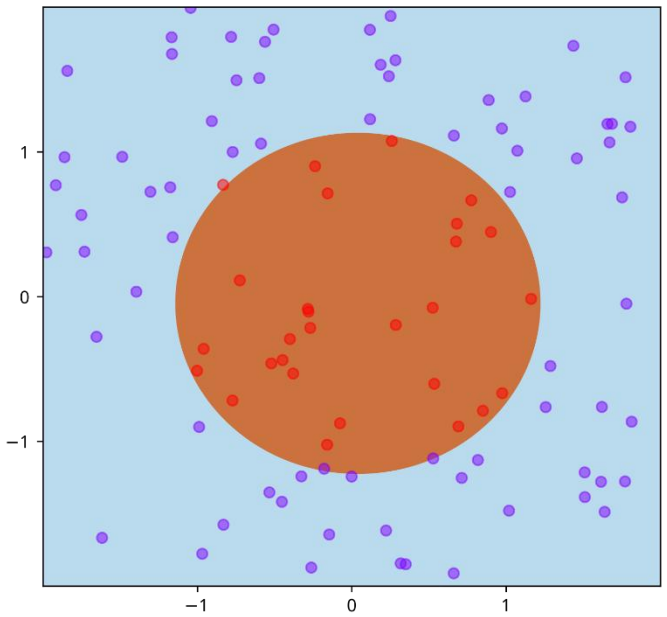
Map data to higher dimensionality!

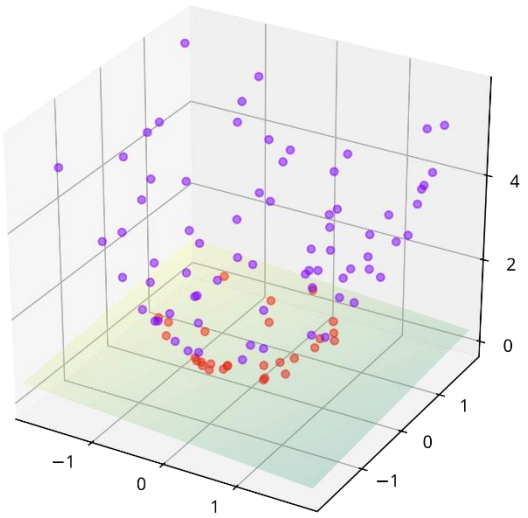
$$x \to \phi(x)$$

Linear model becomes:

$$f(x) = \mathrm{sgn}(w^T \phi(x) - b)$$

But: Not practical to calculate $\phi(x)$ explicitly (combinatorial growth of feature space up to infinite-dimensional!)





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Support Vector Classifier (SVC): The kernel trick

SVM solution: Optimal vector of linear combination of training data

$$w = \sum_{i=1}^{n} \alpha_i x_i y_i$$

In the **dual form**, i.e. comparing x_i and x_j, when we want to minimize $\|w\|^2$:

$$\|w\|^2 = \left(\sum_{i=1}^{n} \alpha_i x_i y_i\right)^{\mathrm{T}} \left(\sum_{j=1}^{n} \alpha_j x_j y_j\right) = \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j x_i x_j y_i y_j$$

Which **only depends on dot product** (between pair of samples): $\boldsymbol{x_i^{\mathrm{T}} x_j}$

…which is a scalar and can be replaced with a **kernel:** $\boldsymbol{K(x_i, x_j)}$

Support Vector Classifier (SVC): The kernel trick

The **kernel** $\boldsymbol{K}(\boldsymbol{x_i}, \boldsymbol{x_j})$ corresponds to a similarity function and delivers a scalar!

Instead of mapping $x \rightarrow \phi(x)$ and computing explicitly all $f(x) = \mathrm{sgn}(w^T \phi(x))$

In the dual form, we compute: $\boldsymbol{K}(\boldsymbol{x_i}, \boldsymbol{x_j}) = \boldsymbol{\phi}(\boldsymbol{x_i})^{\boldsymbol{T}} \boldsymbol{\phi}(\boldsymbol{x_j})$

…dot product = scalar
…no need to calculate $\phi(x)$!
…very efficient in huge feature spaces!

…hence, once trained, the classifier becomes, e.g.: $f(x) = \mathrm{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i \, \boldsymbol{K}(\boldsymbol{x_i}, \boldsymbol{x_j}) - b\right)$

The kernel trick

1) Consider the following minimal dataset:
$x_1 = -2, y_1 = -1$ (class A)
$x_2 = -1, y_2 = -1$ (class B)
$x_3 = 1, y_3 = 1$ (class B)
$x_4 = 2, y_4 = 1$ (class A)

— Plot the data and see if this can be separated in 1D

— Map data using $\phi(x) = (x, x^2)$

— Plot transformed data

— What kernel corresponds to this mapping (assume two points x and z)?

2) Imagine the following kernel $K(x, z) = (1 + x^2 z^2)$. What would be the corresponding feature map (how many dimensions)?

Common Kernels

- Linear kernel (special case): $\boldsymbol{K}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j$

- Polynomial kernel): $\boldsymbol{K}(\boldsymbol{x}_i, \boldsymbol{x}_j) = (\boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j + \boldsymbol{c})^{\boldsymbol{d}}$

- Radial Basis Function (RBF, Gaussian) kernel:
  - $\boldsymbol{K}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \mathbf{exp}\left(-\boldsymbol{\gamma} \cdot \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^{\mathbf{2}}\right)$
  - Most popular for non-linear data
  - Draws smooth, flexible decision boundaries

- Sigmoid kernel: $\boldsymbol{K}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \mathbf{tanh}(\boldsymbol{\beta} \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j + \boldsymbol{\theta})$

Try: https://greitemann.dev/svm-demo

SVC for Breast Cancer set

1) Run a Classification using a SVC with linear kernel on the Wisconsin Breast Cancer dataset (*SVC_breastcancer.ipynb*)
2) Compare different kernels (linear, RBF) and other models on a reduced dataset (more difficult scenario)

In both tasks,

— Use Pipelines

— Perform a GridSearchCV

— Calculate a confusion matrix

Outlook

Regression models based on the kernel trick:

— Support Vector Regressor (SVR)

— Kernel Ridge regression (KRR)

Both use the same kernel trick as SVC, but

— Minimise squared loss (KRR) or ε-insensitive loss (SVR)

— encourage smooth, nonlinear function fitting for continuous data

— no margin concept (except in SVR where a “tube” is used)

Will return to them a when we talk about linear regression.